

Report: root-to-shoot ratio

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1 Summary

Data Exploration

- Root mass slightly more right-skewed than sprout mass
- Sprout-root-mass-ratio shows stronger peak around 0.5 - 0.7
- High correlation between sprout- and root-mass ($r = 0.76$)

Marginal Distribution

- Gamma fits data better than Normal, support with Kullback-Leibler-Divergence

Ratio Modeling

- The theoretical model doesn't match the empirical distribution of the ratios
- Supports statistical dependency of root- and sprout-mass

Interpretation

- The growth of root and sprout might be statistical dependent (with statistical power)
- Since the theoretical model under estimates the peak, one could argue that this ratio range might be a optima for breeding
- Since I don't have much information about the data, I cannot interpret much about the biology

2 Data Exploration

Data Distribution

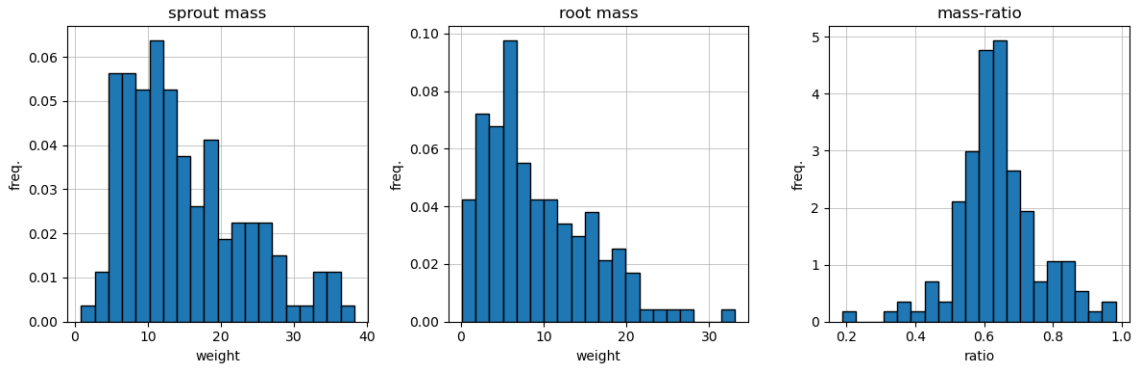


Fig. 1: Exploitative analysis. Left: Sprout mass data distribution. Middle: Root mass data distribution. Right: Mass-Ratio distribution.

Masses

- The masses of sprouts and roots tend to be slightly right skewed (rechtsschief)
- $Weights \in \mathbb{R}_+$
- Just a normal distribution might now be appropriate, because its highly likely it will extend into the negatives and doesn't catch the skewness
- A Gamma might fits the marginal distribution better, because: $X \in \mathbb{R}_+$

Ratio

- The skweness seem to not be present here
- $Ratio \in [0, 1]$
- Intuitivly (and by their definition) a beta might fit best here

Data Correlation

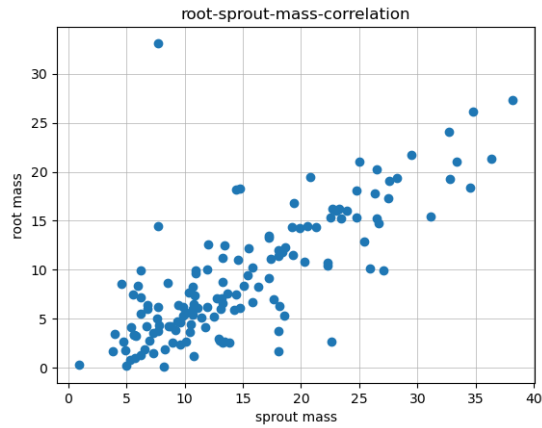


Fig. 2: Scatter plot of root and sprout mass.

- Hight correlation between the masses ($r \approx 0.76$)
- Implying a statistical dependency between both parameters

3 Marginal Distribution

Gamma fits data better then a Normal

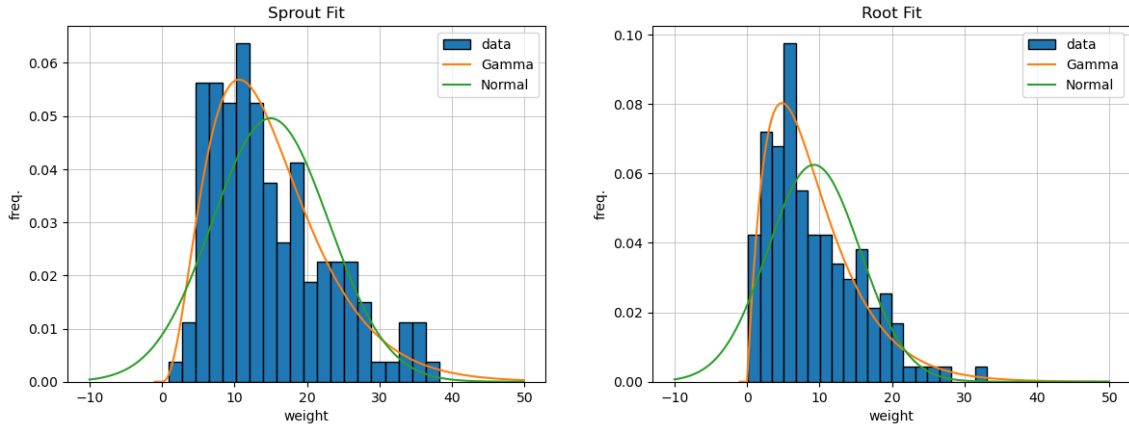


Fig. 3: Continuous Curves of estimated Distributions

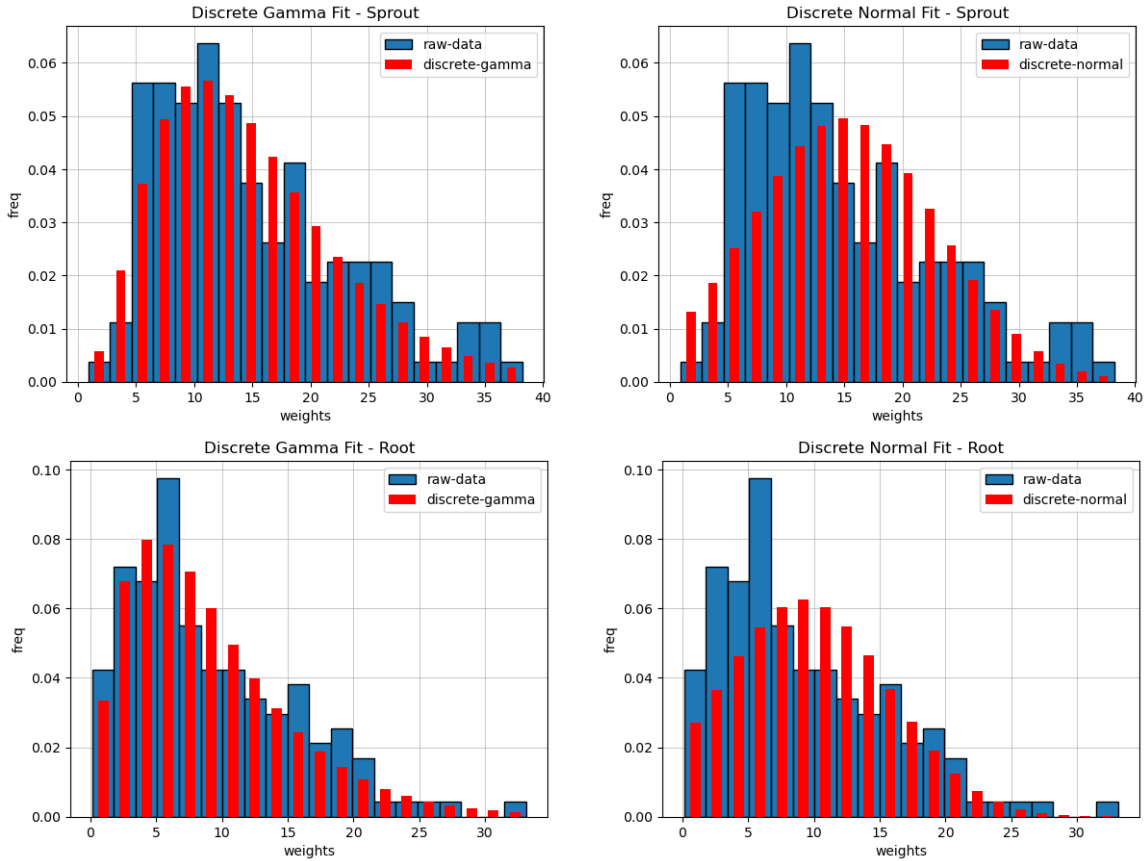


Fig. 4: Discrete bars of estimated Distributions

- Just by visually inspecting the fitting, I would argue that the Gamma fits X_{sprout} as well as X_{root} considerably better then a Normal
- A gamma distribution can model the skewness of data
- Also a normal extends into negatives numbers, which the data clearly is not showing

Estimated parameters from Gamma and Normal (sprout and root)

	α	β	μ	σ
Sprout	3.46	0.231	14.97	8.05
Root	2.09	0.227	9.23	6.38

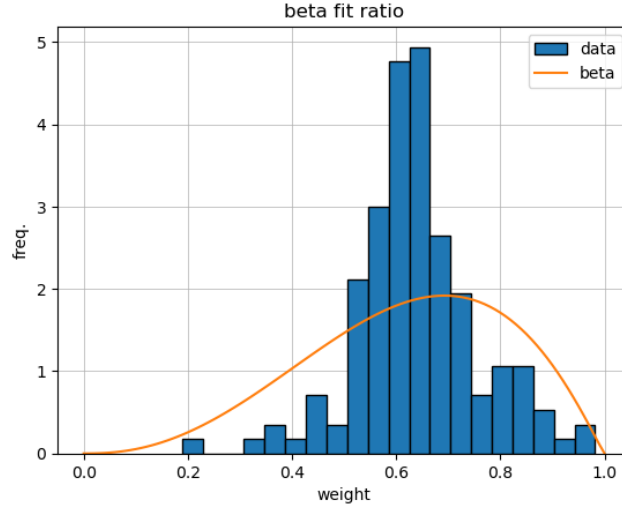


Fig. 5: Theoretical Beta curve fitting.

Further Evaluation using Kullback-Leibler-Divergence

Sprout

- $KLD_{Gamma} = 0.0363$
- $KLD_{Normal} = 0.0936$

Root

- $KLD_{Gamma} = 0.0273$
- $KLD_{Normal} = 0.1159$

KLD support

- KLD clearly supports my previous statement, that Gamma captures data much better, therefore is favorable

Note on the gamma modelling

- One can notice that: $\beta_{root} \approx \beta_{sprout}$
- This supports the theoretical distribution for the ratio transformation later
- But also that: $\alpha_{root} < \alpha_{sprout}$, implying that the skewness of the root distribution is stronger !

4 Ratio-Modelling using Beta Distribution

Root-To-Shoot Ratio

Theoretically

$X_1 \sim Ga(x_1 | \alpha_1, \beta)$ Shown in previous section

$X_2 \sim Ga(x_2 | \alpha_2, \beta)$ Shown in previous section

$Y = \frac{X_1}{X_1 + X_2}$ Transformation

$Y \sim Be(y | \alpha_1, \alpha_2)$ Under the assumption that $\beta_{root} = \beta_{sprout} = \beta$ and $X_1 \perp\!\!\!\perp X_2$ (independent).

- One can clearly see that the theoretically derived beta distribution doesn't fit the empirical ratios

5 Interpretation

- The poor fit quality of the estimated beta distribution is an indicator that both masses are statistically dependent and further supports the correlation
- This validates, that the correlation breaks the independency assumption of the theoretical beta
- One could interpret that this peak is the result of stronger dependency (also correlation) of those seedlings masses
- Unfortunately, I don't have too much information about the data, especially I don't know if certain seedlings perform better than other (for breeding purpose)
- I could argue that this ratio range (0.5 - 0.7) might be a optimal target for breeding selection (with statistical power)